

International Islamic University Chittagong (IIUC)

Faculty of Shariah and Islamic Studies (FSIS)
Department Of Science of Hadith and Islamic Studies (SHIS)
Semester End Examination, Spring-2026

Course Code: BSE-2301
Full Marks: 50

Course Title: Basic Sciences
Time: 2 Hours 30 Minutes

[Answer any Five questions from the following including at least one from each part (Part A, Part B, Part C).
Figures on right margin indicate marks.]

Bloom's Levels of the Questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

Serial No	Questions		Marks	Bloom's Level	COs
Part A: Physics					
1	a	Convert the following: 1. 0.75 km into cm. 2. 2.5 hours into seconds. 3. 4500 cm ³ into m ³ .	3	Apply	CO ₂
	b	A rectangular metal piece has a mass of 540 g and a volume of 200 cm ³ . Find its density in g/cm ³ and kg/m ³ .	2	Apply	CO ₂
	c	State two situations where the work done by a force is zero. Explain briefly.	3	Understand	CO ₁
	d	Show that force is a derived quantity by expressing its SI unit in base units.	2	Understand	CO ₁
2	a	A student walks 400 m North, then 150 m South, and finally 250 m North. Find the total distance travelled and displacement.	3	Apply	CO ₂
	b	Differentiate between speed and velocity. Give one practical example of each type.	3	Understand	CO ₁
	c	Find the kinetic energy of a 3 kg object moving with a speed of 8 m/s.	2	Apply	CO ₂
	d	A 4 kg object is kept at a height of 15 m. Calculate its potential energy.	2	Apply	CO ₂
Part B: Chemistry					
3	a	What is an element? Differentiate between an element and a compound.	3	Understand	CO ₁
	b	What is valency? Find the valency of the following: i) Calcium ii) Oxygen iii) Chlorine iv) Aluminum	4	Apply	CO ₂
	c	Briefly describe ionic and covalent bonds with suitable examples. Write two differences between ionic and covalent bonds.	3	Understand	CO ₁

Serial No	Questions		Marks	Bloom's Level	COs
OR	a	What are atomic number and mass number?	2	Remember	CO ₁
	b	Write the symbols of the following elements: i) Zinc ii) Bromine iii) Nickel iv) Tin	2	Remember	CO ₁
	c	For $^{35}_{17}\text{Cl}^-$, find the mass number, atomic number, charge, number of protons, number of electrons, and number of neutrons.	3	Apply	CO ₂
	d	Identify whether the following pairs are isotopes, isotones, or isobars: 1. $^{12}_6\text{C}$ and $^{14}_6\text{C}$ 2. $^{14}_6\text{C}$ and $^{14}_7\text{N}$	3	Analyze	CO ₂
Part C: Biology					
4	a	What are nutrients? Mention the characteristics of a balanced diet.	3	Understand	CO ₁
	b	Name the major food groups. Write one important function and one food source of each group.	3	Apply	CO ₂
	c	Explain the importance of a food pyramid in maintaining a healthy diet. Draw a simple food pyramid and label its levels.	4	Apply	CO ₂
5	a	What is an ecosystem? Explain how producers, consumers, and decomposers depend on one another in an ecosystem.	4	Apply	CO ₂
	b	In a pond ecosystem, identify the producers, primary consumers, secondary consumers, and decomposers. Draw a simple diagram to show their relationship.	3	Create	CO ₁
	c	A food item contains 70 g carbohydrate, 12 g protein, and 8 g fat per 100 g. Calculate the total energy supplied by 100 g of the food. <i>Use: Carbohydrate = 4 kcal/g, Protein = 4 kcal/g, Fat = 9 kcal/g.</i>	3	Understand	CO ₁
OR	a	Why does night blindness occur? Which nutrient deficiency causes it? Mention two food sources that can prevent this deficiency.	3	Remember	CO ₁
	b	A student has a body weight of 72 kg and a height of 1.68 m. 1. Calculate the BMI of the student. 2. State whether the student is underweight, normal, overweight, or obese.	4	Analyze	CO ₂
	c	What are plant growth regulators? Write the functions of any three plant hormones.	3	Understand	CO ₁

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Final Examination, Autumn-2025

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Full marks: 50

Course Title: Basic Sciences

Time: 2 hours 30 minutes

[Answer any **Five** questions from the followings including at least **one** from each part (part A, part B, part C). Figures on right margin indicate marks.]

Bloom's Levels of the questions						
Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember

Serial No	Questions		Marks	Bloom's level	CLOs
Part A: Physics					
1	a	A stone has mass 250 g and volume 100 cm ³ . Find density in g/cm ³ and kg/m ³ .	2	Apply	COL-2
	b	What is the difference between distance and displacement? Give an example.	2	Analyze	COL-1
	c	If you travel 5 km North, then 3 km East, then 5 km South, what is your displacement from the starting point?	3	Remember	
	d	What is the kinetic energy of a 4 kg object moving at 3 m/s?	3	Create	COL-2
2	a	A box measures 30 cm by 20 cm by 10 m. Find its volume in m ³ .	2	Apply	COL-2
	b	At what height does a 3 kg object have 150 J of potential energy? ($g=10 \text{ m/s}^2$)	3	Analyze	COL-1
	c	What are three systems of units of measurement and write the 7 basic units of measurement?	3	Remember	
	d	Show that power is a derived quantity.	2	Create	COL-2
3	a	What is the difference between mass and weight? Write the SI unit of each.	3	Apply	COL-1
	b	If a ball of mass 2 kg is falling from a height of 10 m with a velocity of 5 m/s. What's its total energy?	4	Apply	COL-1
	c	A pump lifts 500 kg of water to a height of 20 m in 2 minutes. What is the power of the pump? ($g = 10 \text{ m/s}^2$)	3	Understand	COL-2
Part B: Chemistry					
4	a	What is element? Differentiate Between Atom and Molecule.	2	Understand	COL-1
	b	What is valency? Find the valency of. a) Magnesium b) Sulfur	3	Remember	COL-2
	c	Write the chemical symbols and proton numbers for. a) Potassium	2	Analyze	COL-1

		b) Lead c) Sodium d) Gold			
	d	Write the full electron configuration for Iron (Fe, Z=26) and identify its group number.	3	Evaluation	CO ₃
5	a	Write the set of four quantum numbers for the last electron of Potassium.	4	Create	CO ₃
	b	What is chemical bond? Write down the different types of chemical bond?	3	Remember	CO ₃
	c	How many periods and groups are in the periodic table? Write the formula for: a) Water b) Carbon dioxide c) Sodium chloride	3	Apply	CO ₃

Part C: Biology

6	a	What is the difference between a "producer" and a "consumer" in an ecosystem? Give two examples of each.	3	Understand	CO ₃
	b	What are food groups? Why does night blindness occur?	3	Apply	CO ₃
	c	What is a "food chain"? Write an example of a simple 4-step food chain starting from grass.	4	Apply	CO ₃
7	a	A boy is 15 years old, weighs 45 kg, and is 1.6 meters tall. Calculate his Body Mass Index (BMI) and state which category he falls into (Underweight, Normal, Overweight).	4	Apply	CO ₃
	b	What is the main job of the "stomata" on a leaf? When do they usually open and close?	3	Create	CO ₃
	c	If 20 gm chira contains 15.4 grams carbohydrate (77%), 1.32 grams protein (6.6%) and 0.24 grams fat. What is the calorie of 1 KG chira?	3	Understand	CO ₃

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Faculty of Shariah and Islamic Studies (FSIS)

Final Examination, Spring-2025

Course Code: BSE-2301

Full marks: 50

Course Title: Basic Sciences

Time: 2 hours 30 minutes

[Answer any **Five** questions from the followings including at least **one** from each part (part A, part B, part C). Figures on right margin indicate marks.]

Bloom's Levels of the questions

Bloom's Meaning	Create	Evaluate	Analyze	Apply	Understand	Remember
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Serial No	Questions		Marks	Bloom's level	CLOs
Part A: Physics					
1	a	Convert the following: 1) 350 cm into km. 2) 720 minutes into days. 3) 100 cm 500 mm into m.	3	Apply	COL-2
	b	If mass is 500 grams and volume 1 m^3 find density in kg/m^3	2	Analyze	COL-1
	c	What are three systems of units of measurement and write the 7 basic units of measurement?	3	Remember	
	d	Show that work is a derived quantity.	2	Create	COL-2
2	a	A person pulls a cart with 100 N force at 60° angle over 10 m. Find work done.	3	Apply	COL-1
	b	If a ball of mass 2 kg is falling from a height of 10 m with a velocity of 5 m/s. What's its total energy?	3	Apply	COL-1
	c	Define Work. Sabbir's weight is 50 kg, and he climbs to the top of a 10-story building. Each floor is 3 meters high. How much work has he done? It takes 5 minutes to get to the 10th floor, find his power?	4	Understand	COL-2
Part B: Chemistry					
3	a	What is element? Differentiate Between Atom and Molecule.	2	Understand	COL-1
	b	Write the symbol of the following elements. 1. Silver 2. Iron 3. Mercury 4. Tungsten	2	Remember	COL-2
	c	Find out the mass number, atomic number, charge, proton number, electrons and neutrons number for $^{27}_{13}\text{Al}^+$	3	Analyze	COL-1
	d	What is isotope? identified $^{14}_6\text{C}$, $^{16}_8\text{O}$ and $^{24}_{12}\text{Mg}$ are isotone or not?	3	Evaluation	COL-2

4	a	What is Valence electron? What is the Valence and Valency of $^{16}_8O$, $^{23}_{11}Na$, $^{35}_{17}Cl$.	4	Create	CO
	b	Briefly describe ionic and covalent bonds with suitable examples. Write 2 differences between ionic and covalent bonds.	3	Remember	COL-1
	c	What is Bohr's atomic model? How is it better than Rutherford's model?	3	Apply	COL-2
Part C: Biology					
5	a	What is nutrition? Write down the characteristics of balanced diet.	3	Understand	COL-1
	b	Our body needs different kinds of food to stay strong. What are the main food groups and what does each one do? Write their names and roles.	3	Apply	COL-2
	c	Draw an ideal food pyramid? Explain the concept of transpiration.	4	Apply	COL-2
6	a	Green plants make their own food using sunlight. What is this process called? Explain it step by step by naming and describing its main phases.	4	Apply	COL-2
	b	What is biomass? Describe the role of producer, consumer and the decomposers in maintaining the ecosystem of a pond with diagram.	3	Create	COL-1
	c	If 20 gm chira contains 15.4 grams carbohydrate (77%), 1.32 grams protein (6.6%) and 0.24 grams fat. What is the calorie of 1 KG chira?	3	Understand	COL-2
7	a	What is ecosystem? Write down the three types of transpiration name.	3	Remember	COL-1
	b	A woman has a height of 155 cm and a weight of 62 kg. Her age is 33. 1) Calculate her BMI. 2) Calculate her BMR.	4	Analyze	COL-1
	c	What are plant growth regulators? What is the function of abscisic acid?	3	Understand	COL-1

International Islamic University Chittagong

Faculty of Shari'ah and Islamic Studies

Department of Science of Hadith and Islamic Studies

Course Code: BSE-2301, Course Title: Basic Sciences

Final Exam, Spring-2024

Total Marks: 50

Time: 2 hour 30 minutes

[Answer any five questions from the followings including at least one from each parts (part A, part B and part C). Figures on right margin indicate marks.]

Part A: Biology

- 1.(a) Describe the nutrient elements consumed by plants? Write the source and role of some nutrient elements. 5
- 1.(b) Define photosynthesis. Explain the process of photosynthesis with figure. 5
- 2.(a) Write the characteristics of balance diet? Explain some deficiency symptoms associated with nutrients. 5
- 2.(b) Define Biomass. Explain how 'Transpiration is a necessary evil'. 5
- 3.(a) Explain ecosystem and its components? 5
- 3.(b) Describe the role of Light and Chlorophyll in Photosynthesis. 5

Part B: Physics

- 4.(a) Convert the following lengths to meters, times to seconds and mass to grams
i) 10 centimeters, ii) 36 hours, iii) 50 kilograms 3
 - 4.(b) Calculate the density of water if it has a mass 1160kg and volume of 3 m^3 . 3
 - 4.(c) Describe Necessity of measurement in our daily life. 4
 - 5.(a) What is the SI unit of Energy? Calculate the kinetic energy of a 200kg object moving at a speed of 15 m/s. 5
 - 5.(b) A garage hoist lifts a truck up 2 meters above the ground in 15 seconds. Find the power delivered to the truck. (1000 kg as the mass of the truck) 5
- Or,**
What do you mean by work is 1 watt? A force is applied on a stationary object of mass 10kg for 10s, what is the kinetic energy of the object?

Part C: Chemistry

- 6.(a) What are the system of writing symbol of element? Define three state of matter and give an example of each? 5
 - 6.(b) What is the difference between an atom and a molecule? Discuss the limitation of Rutherford's model. 5
- Or,**
Define inter molecular attraction. Discuss the success of Bohr's model.
- 7.(a) Write the symbol of hydrogen, oxygen, carbon and potassium? Discuss the limitation of Bohr's model? 5
 - 7.(b) What is the valence of chlorine (Cl(17)) and hydrogen (H(1))? Draw the formation of a chemical bond using hydrogen and chlorine atoms combining to form a hydrogen chloride molecule (HCl). 5